

Milestone 4: Testing & Management

1. Test Plan

1.1 Introduction

The purpose of testing is to ensure that the Student Mental Health Early Warning System functions correctly, meets user requirements, and performs efficiently under different conditions. Testing helps identify bugs, improve reliability, and validate the accuracy of risk classification and alert mechanisms.

1.2 Types of Testing

A. Unit Testing

Unit testing checks individual modules or components independently.

Modules Tested

- Student Registration Module
- Login Authentication Module
- Assessment & Risk Scoring Module
- Counselor Alert Module
- Mood Tracking Module
- Report Generation Module

Example

The risk scoring module is tested to verify that a PHQ-9 score of 15 or above correctly classifies the student as High risk and triggers an alert.

B. Integration Testing

Integration testing ensures that multiple modules work correctly together.

Integration Areas

- Assessment module connected with Risk Engine
- Risk Engine connected with Alert Service
- Alert Service connected with Notification Service
- Report module connected with Assessment and Mood databases

Example

When a student submits a High-risk assessment, the alert service should automatically notify the assigned counselor within 60 seconds.

C. System Testing

System testing validates the complete application as a whole.

Features Checked

- Complete workflow: registration → assessment → risk scoring → counselor alert
- User interface behavior across web and mobile
- System performance under concurrent users
- Data encryption and secure storage
- Notification accuracy and delivery timing

1.3 Sample Test Cases

Test Case ID	Module	Test Description	Input	Expected Output	Result
TC-01	Login	Verify valid login	Correct email/password	User redirected to dashboard	Pass
TC-02	Login	Verify invalid login	Wrong password	Error message displayed	Pass
TC-03	Assessment	High risk classification	PHQ-9 score = 18	Risk = High, counselor alerted	Pass
TC-04	Assessment	Low risk classification	PHQ-9 score = 2	Risk = Low, no alert triggered	Pass
TC-05	Mood Tracker	Log mood entry	Mood score = 3	Entry saved, chart updated	Pass
TC-06	Alert	Counselor alert delivery	High-risk student detected	Counselor notified within 60 seconds	Pass
TC-07	Report	Generate wellness report	Date range: 1 month	PDF report generated with charts	Pass
TC-08	Anonymous	Submit anonymous report	Concern text entered	Report forwarded, identity stripped	Pass

2. Version Control Usage

2.1 Introduction

GitHub was used as the Version Control System (VCS) for managing the SMHEWS project documentation and design files. The repository stores UML diagrams, reports, requirement documents, and planning files.

2.2 Benefits of Using GitHub

- Maintains complete project history
- Enables collaboration among team members
- Tracks all modifications and bug fixes
- Allows rollback to previous stable versions
- Supports branch management and merging

2.3 GitHub Workflow

The following workflow was followed during development:

1. Repository created on GitHub
2. Team members cloned the repository
3. Separate branches created for each milestone/feature
4. Changes committed with descriptive messages after each session
5. Pull requests used for code review before merging
6. Final version maintained in the main branch

2.4 GitHub Repository Link

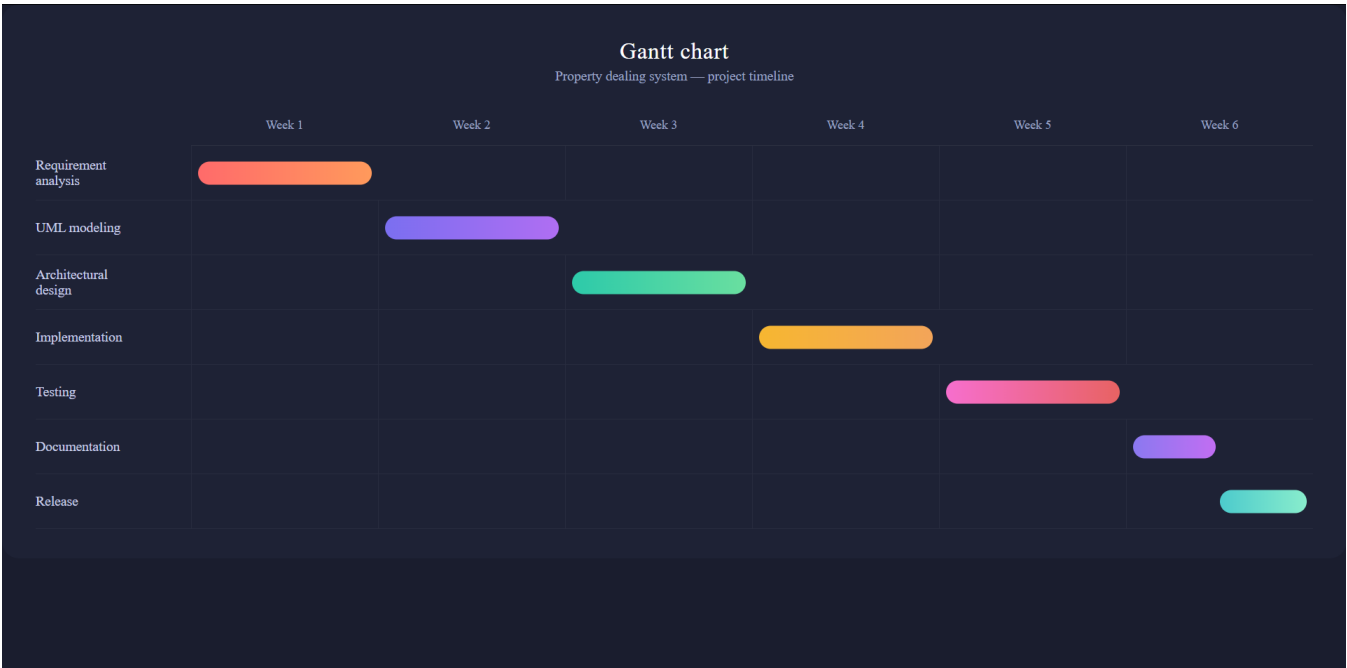
https://github.com/ammaarr14/Student_Mental_Health_Warning_System

3. Project Planning Report

3.1 Gantt Chart Description

A Gantt chart was prepared to manage project scheduling and monitor progress across 6 weeks.

Project Timeline



3.2 Risk Analysis

Risk	Impact	Mitigation Strategy
Requirement changes	Delay in project	Frequent review meetings with supervisor
Data privacy breach	Legal/ethical issues	AES-256 encryption + strict access control
Team communication issues	Reduced productivity	Weekly Scrum standups + Slack coordination
Inaccurate risk classification	Harm to students	Validate scoring against clinical PHQ-9 standards
Database failure	Data loss	Automated daily backups to cloud storage
Time management issues	Late submission	Gantt chart tracking with milestone deadlines

3.3 Team Roles and Responsibilities

Team Member	Role	Responsibilities
Muhammad Talha Saeed	Project Manager	Managing schedule, coordination, and milestone tracking
Muhammad Ammaar Abdullah	System Analyst	Requirement analysis, use case documentation
Muhammad Saad Shahzad	Backend Developer	Risk engine, database design, API development
Afnan	Frontend Developer	UI design, dashboard implementation, mobile responsiveness
(All Members)	Tester	Unit testing, integration testing, and bug reporting